# SINGAPORE SCHOOL

## CONTINUAL ASSESSMENT 1

**Secondary 1**

**Computer Science (Part 1)**

**2026-27**

**Date: ___ / ___ / ___** **Name: _____**

**Instructions for students:**

- **Start answering the questions on time.**
- **Follow all the instructions carefully.**
- **Answer all questions.**
- **Revise**

| | |
|---|---|
| **CA 1 Marks (Part – 1)** | **25** |
| **CW** | **05** |
| **Total Marks Obtained** | **30** |

**1. Choose the correct answer for each of the following multiple-choice questions. (1 x 10 = 10)**

(a) What is the base of the binary number system?
A) Base 2 B) Base 8 C) Base 10 D) Base 16
**Answer: [ ]**

(b) Convert the 4-bit binary number 1101 into decimal (denary):
A) 11 B) 12 C) 13 D) 14
**Answer: [ ]**

(c) Convert the decimal (denary) number 25 into an 8-bit binary number:
A) 00011001 B) 00010101 C) 00110001 D) 00011100
**Answer: [ ]**

(d) How many binary bits are represented by a single hexadecimal digit?
A) 2 bits B) 4 bits (1 nibble) C) 8 bits (1 byte) D) 16 bits
**Answer: [ ]**

(e) In the hexadecimal number system, what decimal value does the letter 'E' represent?
A) 12 B) 13 C) 14 D) 15
**Answer: [ ]**

(f) Convert the binary number 1010 1111 into hexadecimal:
A) 9F B) AF C) B7 D) AE
**Answer: [ ]**

(g) Convert the hexadecimal number 3C into an 8-bit binary number:
A) 0011 1100 B) 0011 1010 C) 0110 1100 D) 0011 1101
**Answer: [ ]**

(h) In binary addition, what is the result of 1 + 1?
A) 2 with carry 0 B) 1 with carry 1 C) 0 with carry 1 D) 0 with carry 0
**Answer: [ ]**

(i) In binary addition, what is the result of adding three ones (1 + 1 + 1)?
A) 1 with carry 1 B) 0 with carry 1 C) 1 with carry 0 D) 3 with carry 0
**Answer: [ ]**

(j) What is the result of adding the 4-bit binary numbers 0101 + 0011?
A) 0111 B) 1000 C) 1001 D) 1010
**Answer: [ ]**

**2. Convert each of the following 8-bit binary numbers into decimal (denary). Show your working. (1 x 3 = 3)**

(a) 0010 1101

*Working & Answer Box:*

**Answer:** _____

(b) 0101 0110

*Working & Answer Box:*

**Answer:** _____

(c) 1001 1011

*Working & Answer Box:*

**Answer:** _____

**3. Convert each of the following decimal (denary) numbers into 8-bit binary. Show your working. (1 x 3 = 3)**

(a) 42

*Working & Answer Box:*

**Answer:** _____

(b) 99

*Working & Answer Box:*

**Answer:** _____

(c) 205

*Working & Answer Box:*

**Answer:** _____

**4. Convert each of the following 8-bit binary numbers into hexadecimal. (1 x 3 = 3)**

(a) 0100 1101

*Working & Answer Box:*

**Answer:** _____

(b) 1011 0111

*Working & Answer Box:*

**Answer:** _____

(c) 1110 1010

*Working & Answer Box:*

**Answer:** _____

**5. Convert each of the following hexadecimal values into 8-bit binary. (1 x 3 = 3)**

(a) 59

*Working & Answer Box:*

**Answer:** _____

(b) A6

*Working & Answer Box:*

**Answer:** _____

(c) F8

*Working & Answer Box:*

**Answer:** _____

**6. Perform binary addition on each of the following pairs of 8-bit binary numbers. Show all carry bits clearly. (1.5 x 2 = 3)**

(a) 0010 0111 + 0011 1010

*Working & Answer Box:*

**Answer:** _____

(b) 0101 1101 + 0010 1110

*Working & Answer Box:*

**Answer:** _____